Oppenheimer and Serber had no idea that Landau's paper was a desperate attempt to avoid prison and possible death, so on 1 September 1938, as Landau languished in Butyrskaya Prison, they submitted their critique of him to the *Physical Review.* Since Landau was a great enough physicist to take the heat, they said quite frankly: "An estimate of Landau . . . led to the value 0.001 solar masses for the limiting [minimum] mass [of a neutron core]. This figure appears to be wrong. . . . [Nuclear forces] of the often assumed spin exchange type preclude the existence of a [neutron] core for stars with masses comparable to that of the Sun."

Landau's neutron cores and Zwicky's neutron stars are really the same thing. A neutron core is nothing but a neutron star that happens, somehow, to find itself inside a normal star. To Oppenheimer this must have been clear, and now that he had begun to think about neutron stars, he was drawn inexorably to the issue that Zwicky should have tackled but could not: What, precisely, is the fate of massive stars when they exhaust the nuclear fuel that, according to Bethe and Critchfield, keeps them hot? Which corpses will they create: white dwarfs? neutron stars? black holes? others?

Chandrasekhar's calculations had shown unequivocally that stars less massive than 1.4 Suns must become white dwarfs. Zwicky was speculating loudly that at least some stars more massive than 1.4 Suns will implode to form neutron stars, and in the process generate supernovae. Might Zwicky be right? And will all massive stars die this way, thus saving the Universe from black holes?

One of Oppenheimer's great strengths as a theorist was an unerring ability to look at a complicated problem and strip away the complications until he found the central issue that controlled it. Several years later, this talent would contribute to Oppenheimer's brilliance as the leader of the American atomic bomb project. Now, in his struggle to understand stellar death, it told him to ignore all the complications that Zwicky was trumpeting about—the details of the stellar implosion, the transformation of normal matter into neutron matter, the release of enormous energy and its possible powering of supernovae and cosmic rays. All this was irrelevant to the issue of the star's *final fate.* The only relevant thing was the maximum mass that a neutron star can have. If neutron stars can be arbitrarily massive (curve *B* in Figure 5.3 above), then black holes can never form. If there is a